Exp No:2) Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
df = pd.DataFrame({
   'Date': ['01-01-2023', '02-01-2023','03-01-2023','04-01-2023','05-01-2023',],
   'Product': ['Product A', 'Product B','Product A', 'Product C','Product B',],
   'Sales': [200,150,220,300,180,],
   'Quantity': [4,3,5,6,4,],
   'Region': ['North','South','North','East','West',]
})
print(df.head())
print(df.isnull().sum())
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
print(df.describe())
product_summary = df.groupby('Product').agg({
   'Sales': 'sum',
   'Quantity': 'sum'
}).reset_index()
print(product_summary)
```

```
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date')['Sales'].sum().reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'],
marker='o')
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.xticks(rotation=45)
plt.show()
pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
    fill_value=0
)
print(pivot_table)
```

```
correlation_matrix = df.corr(numeric_only=True)

print(correlation_matrix)

plt.figure(figsize=(8, 6))

plt.imshow(correlation_matrix, cmap='coolwarm')

plt.colorbar()

plt.xticks(range(len(correlation_matrix.columns)),
correlation_matrix.columns)

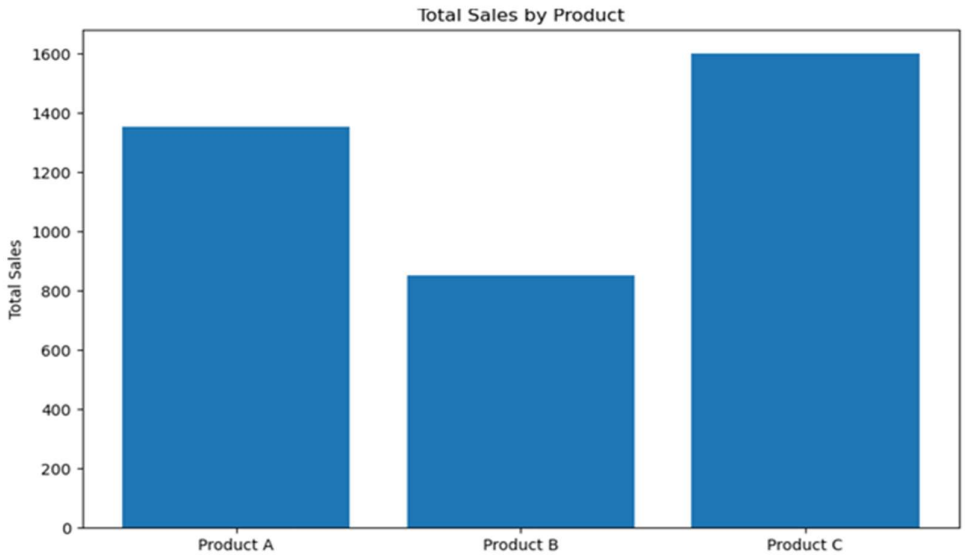
plt.yticks(range(len(correlation_matrix.columns)),
correlation_matrix.columns)

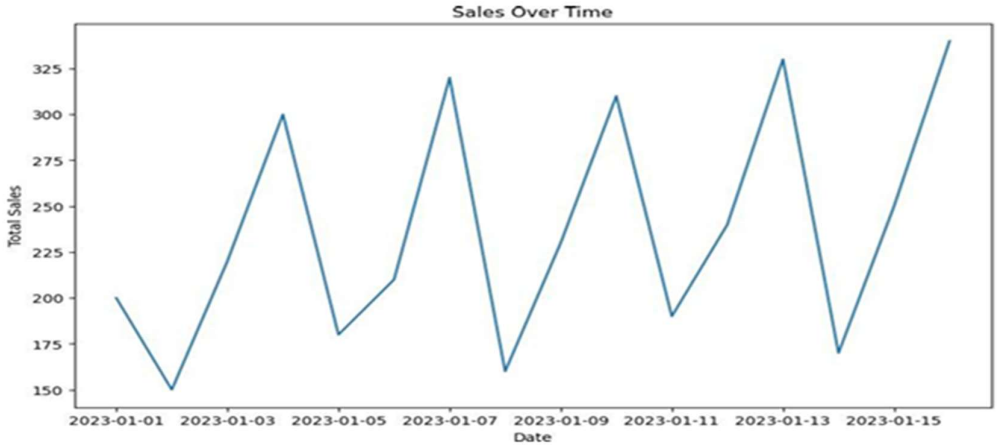
plt.title('Correlation Matrix')

plt.show()
```

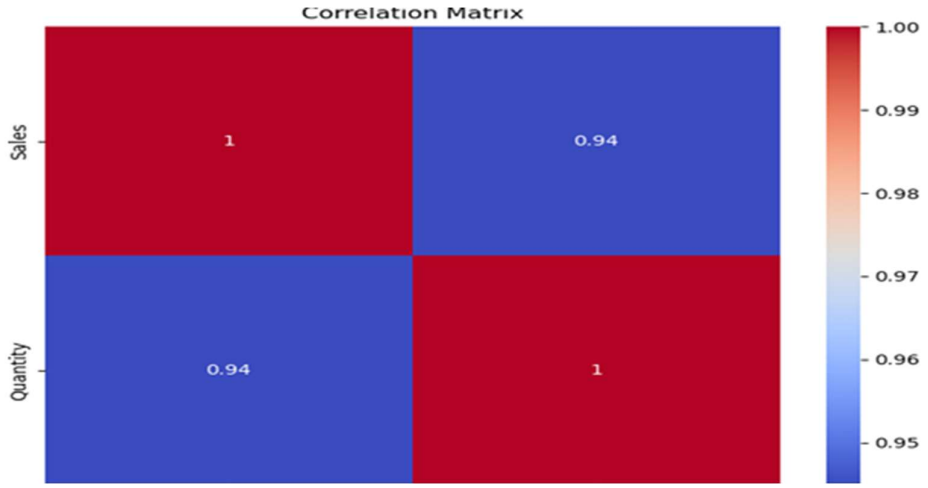
Output:

```
  Date     Product  Sales  Quantity Region
0   2023-01-01  Product A    200         4  North
1   2023-01-02  Product B    150         3  South
2   2023-01-03  Product A    220         5  North
3   2023-01-04  Product C    300         6   East
4   2023-01-05  Product B    180         4   West
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
            Sales   Quantity
count   16.000000  16.000000
mean   237.500000   5.375000
std     64.031242   1.746425
min    150.000000   3.000000
25%    187.500000   4.000000
50%    225.000000   5.500000
75%    302.500000   7.000000
max    340.000000   8.000000
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```






```
Product  Product A  Product B  Product C
Region
East             0          0       1600
North         1350          0          0
South            0        480          0
West             0        370          0
             Sales  Quantity
Sales     1.000000  0.944922
Quantity  0.944922  1.000000
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_9648\511106317.py:49: Futur
eWarning: The default value of numeric_only in DataFrame.corr is deprecated
. In a future version, it will default to False. Select only valid columns
or specify the value of numeric_only to silence this warning.
  correlation_matrix = df.corr()
```



Result:

The data set given in csv format and perform data preprocessing and visualization was analysed and this code was executed successfully.